Supplementary Material

Group-Relative Reinforcement Learning for Blood–Brain-Barrier Molecule Design: What Restrains Reward Hacking, and What Does Not

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All tables below support the main paper. Code, result tables and run histories: https://github.com/JSR996/bbb-gnn-permeability (branch `main`, tag `recomb2027-final`; this document's sources are in `paper/`). Tables S1, S5, S6, S7 and S10 are generated directly from the repository's CSV/JSON files by `make_supp.py`; the others are copied from the repository's documentation (`documentation/*.md`) at the same commit.

1 Classifier

Table S1: Test ROC-AUC per model (mean ± sd over 10 seeds; each seed a different scaffold split). Wang ($n = 55$) and Adenot ($n = 59$) are scaffold-disjoint external holdouts scored by models trained on the indicated corpus. LightGBM uses the same 8 descriptors as the hybrid models.

family	model	BBBP	B3DB	Wang←BBBP	Wang←B3DB	Adenot←BBBP
LightGBM	LightGBM (8 desc.)	0.880 ± 0.038	0.852 ± 0.040	–	–	–
base GNN	gat	0.892 ± 0.038	0.886 ± 0.056	0.833 ± 0.075	0.811 ± 0.037	0.965 ± 0.023
	gcn	0.895 ± 0.037	0.889 ± 0.051	0.843 ± 0.042	0.811 ± 0.036	0.981 ± 0.009
	gin	0.881 ± 0.038	0.883 ± 0.056	0.845 ± 0.049	0.859 ± 0.037	0.984 ± 0.019
	sage	0.889 ± 0.033	0.893 ± 0.051	0.862 ± 0.048	0.841 ± 0.039	0.984 ± 0.029
edge-aware	gat-edge	0.890 ± 0.041	0.888 ± 0.051	0.831 ± 0.031	0.791 ± 0.051	0.967 ± 0.016
	gine	0.886 ± 0.028	0.887 ± 0.059	0.876 ± 0.047	0.845 ± 0.043	0.991 ± 0.011
	sage-edge	0.893 ± 0.031	0.888 ± 0.052	0.850 ± 0.079	0.835 ± 0.034	0.986 ± 0.015
hybrid, GraphNorm	gat	0.908 ± 0.029	0.893 ± 0.058	0.837 ± 0.067	0.814 ± 0.025	0.991 ± 0.008
	gcn	0.908 ± 0.030	0.895 ± 0.053	0.834 ± 0.041	0.824 ± 0.035	0.990 ± 0.004
	gin	0.886 ± 0.038	0.889 ± 0.053	0.844 ± 0.077	0.823 ± 0.036	0.990 ± 0.008
	sage	0.907 ± 0.030	0.900 ± 0.052	0.840 ± 0.052	0.832 ± 0.034	0.990 ± 0.011
hybrid, BatchNorm	gat	0.912 ± 0.024	0.900 ± 0.048	0.829 ± 0.044	0.826 ± 0.033	0.993 ± 0.008
	gcn	0.921 ± 0.032	0.903 ± 0.046	0.850 ± 0.033	0.854 ± 0.032	0.999 ± 0.002
	gin	0.901 ± 0.037	0.905 ± 0.047	0.832 ± 0.062	0.832 ± 0.035	0.998 ± 0.004
	sage	0.916 ± 0.033	0.904 ± 0.040	0.887 ± 0.029	0.855 ± 0.023	0.998 ± 0.003

Table S2: Component effects on ROC-AUC, pooled over 4 operators × 2 datasets × 10 seeds, paired (full table behind Figure 3 of the paper).

component	Δ ROC-AUC	SEM	verdict
wide atom-level readout (shared skeleton)	+0.1304	–	dominant factor
8 RDKit descriptors	+0.0099	0.0019	significant
BatchNorm over GraphNorm	+0.0074	0.0023	significant
descriptors + BatchNorm together	+0.0195	0.0024	significant, additive
BRICS motif graph, intra + atom readout	+0.0064	0.0059	within noise
virtual node	+0.0003	0.0023	within noise
BRICS motif graph, added on top	−0.0042	0.0026	within noise
edge (bond) features	≈ 0	–	within noise
long-range atom connectivity	−0.0194	–	slightly harmful
HRM two-timescale recurrence	−0.0381	0.0059	significant, worse
BRICS motif graph replacing atoms	−0.1161	0.0107	significant, much worse

Table S3: Calibration of the reward-family classifier (hybrid + GraphNorm test folds, 10 seeds pooled, $n = 1970$): Brier 0.1025, ECE 0.0741. Every bin is underconfident, but observed rate is monotone in the score, so ranking is preserved. On Wang: Brier 0.1216, ECE 0.0944 with 4–7 molecules per bin (shape not characterizable).

predicted bin	n	mean predicted	observed rate	gap
0.0–0.1	234	0.048	0.132	+0.084
0.2–0.3	60	0.252	0.600	+0.348
0.4–0.5	66	0.447	0.682	+0.234
0.7–0.8	92	0.756	0.880	+0.124
0.9–1.0	957	0.967	0.975	+0.008

2 Generator: anchors, ablation, screening

Table S4: Anchor interventions beyond the KL sweep of the main text.

intervention	measured	seeds
size gate ($\geq$ 10 heavy atoms), KL 0.3	scaffold fraction $0.769 \pm 0.027 \rightarrow 0.828 \pm 0.004$; undersized molecules 10.4% → 2.1%; cost 0.004 in mean C	3
logit transform of C	scaffold fraction 0.741 → 0.349; policy converged on CCl_4 and freon-12 (CCl_4 reward 2.280, caffeine 2.063)	1
invalid floor $f = -1.0$	scaffold 0.834 against 0.887 at $f = -2$; undersized molecules more than double	1
invalid floor $f = -4.0$	scaffold 0.881 (no change)	1
BRICS KL 3.0	scaffold 0.906 at $C = 0.781$ (KL 1.0 is the knee, not an optimum)	1

Table S5: Gate arm versus arm A on both corpora, computed from the four `screened.csv` files of arms A and gate on each corpus. Survivors pass the screen and are not in the (flattened) training corpus; corpus covered = distinct training molecules reproduced / corpus size.

corpus	pooled A	pooled gate	change	surv. A	surv. gate	change	corpus covered
BBBP ($\tau = 0.615$)	19,219	33,227	+72.9%	6,591	11,329	+71.9%	88.9%→68.4%
B3DB ($\tau = 0.476$)	14,591	17,263	+18.3%	4,166	5,369	+28.9%	46.1%→45.0%

Table S6: Term ablation per seed (200 steps, KL 0.02; "none" is the paired baseline). Δ scaffold is relative to the same-seed baseline. The collapse screen is scaffold fraction < 0.30 (arbitrary).

dropped	seed	scaffold	Δ scaffold	Tanimoto	mean C	collapsed
–	0	0.055	–	0.292	0.991	yes
–	1	0.100	–	0.306	0.982	yes
–	2	0.192	–	0.222	0.980	yes
D	0	0.059	+0.004	0.117	0.983	yes
D	1	0.119	+0.019	0.132	0.982	yes
D	2	0.120	-0.072	0.134	0.982	yes
Q	0	0.128	+0.072	0.367	0.980	yes
Q	1	0.116	+0.016	0.439	0.987	yes
Q	2	0.168	-0.024	0.551	0.985	yes
C	0	0.174	+0.119	0.440	0.723	yes
C	1	0.202	+0.102	0.382	0.823	yes
C	2	0.212	+0.020	0.390	0.812	yes
S	0	0.365	+0.309	0.667	0.980	no
S	1	0.821	+0.721	0.836	0.955	no
S	2	0.441	+0.248	0.678	0.982	no

Table S7: Screening verdict by arm for every pooled generated molecule (mean reward-ensemble *C*). Undersized molecules score highest in every arm.

	arm A		arm B (hybrid-BN reward)		gate arm	
verdict	*n*	mean *C*	*n*	mean *C*	*n*	mean *C*
survivor	6,591	0.854	6,848	0.846	11,329	0.826
in training data	732	0.834	731	0.832	596	0.855
undersized	475	0.929	521	0.928	545	0.928
structural defect	444	0.639	423	0.626	766	0.586
targeted alert	6,328	0.773	6,823	0.771	12,041	0.721
catalogue alert	4,649	0.779	4,736	0.773	7,950	0.750

Table S8: Dominance-and-crowding (NSGA-II-style) reduction versus the linear sum; SELFIES, KL 0.3, 200 steps, four seeds, paired. "seeds" = seeds improved over linear, out of 4. The composite arm keeps the rank order but borrows the linear reward's scale.

metric	linear	NSGA-II	seeds	composite	seeds
TPSA spread ratio	0.410	0.524	4/4	0.519	4/4
log *P* spread ratio	0.683	0.784	4/4	0.785	4/4
MW spread ratio	0.481	0.585	4/4	0.602	4/4
Tanimoto distance	0.887	0.899	4/4	0.899	4/4
KL to reference	0.066	0.047	4/4	0.048	4/4
mean *C*	0.925	0.891	0/4	0.894	0/4
scaffold fraction	0.811	0.780	1/4	0.785	1/4
valid fraction	0.963	0.953	1/4	0.954	2/4
C_{BBBP}	0.924	0.891		0.894	
C_{B3DB}	0.888	0.849		0.848	
instrument gap	+0.036	+0.042	worse 3/4	+0.046	worse 4/4

Table S9: One 1000-step run at the settled configuration (KL 1.0, reward seed 0, policy seed 0); 100-step window means. The apparent early decay is a settling transient: regressing scaffold fraction on step gives slope -2.61×10^{-5} ($p = 0.001$) from step 250 but -1.47×10^{-5} ($p = 0.18$) from 400, -1.48×10^{-5} ($p = 0.30$) from 500 and -1.51×10^{-5} ($p = 0.62$) from 700. One seed, one configuration; any sustained decay is slower than about 0.024 per 1000 steps.

window	scaffold	unique	C	reward	sanitization
0–99	0.835	0.985	0.839	1.702	0.9998
200–299	0.873	0.984	0.878	2.141	0.9997
400–499	0.859	0.982	0.894	2.171	0.9997
600–699	0.861	0.984	0.901	2.195	1.0000
800–899	0.852	0.979	0.904	2.197	1.0000
900–999	0.853	0.979	0.905	2.198	0.9998

Table S10: Settled generator configuration, read from `results/brics_cgate/pseed0/config.json`. The reward weights are annealed linearly over `k_anneal` steps from $(D,C,Q,S) = (1.0,0.2,0.5,0.5)$ to $(0.3,1.5,0.5,0.5)$; with `drop=['C']` the C term is replaced by the gate `min_c`.

setting	value (gate arm, policy seed 0)
steps	250
group_size	64
ppo_epochs	4
n_d	2
lr_g	0.0001
lr_d	0.0001
clip_eps	0.2
kl_coef	1.0
invalid_floor	-2.0
schedule	linear
k_anneal	100
max_len	72
min_heavy_atoms	10
min_c	0.615
drop	['C']
reward_family	ensemble
reward_seed	0
generator	brics
threads_requested	6

3 Reproduction

From a fresh clone of the repository (branch `main`; see `CLAUDE.md` and `documentation/full_pipeline.md`):

```
pip install -r requirements.txt     # requirements.lock pins versions
python -m src.pipeline --list       # what would run / skip
python -m src.pipeline              # classifier -> reward -> generator
                                    #   -> screen -> report
python -m src.train_hybrid --run-all --norm batch
                                    # reported classifier
python -m src.reward                # reward gate; if it fails, stop
python -m src.brics_generator --pretrain   # MLE warm start (required)
python -m src.train_gan --generator brics --steps 250 --threads 6 \
    --seed 0 --reward-seed 0 --kl-coef 1.0 --drop C --min-c 0.615 \
    --warm-start results/generator/bbbp_brics_pretrained.pt \
    --out-dir results/brics_cgate/pseed0
python -m src.postfilter results/brics_cgate/* --from-step 150 \
    --rank-by mpo_gated --out results/candidates_mpo/brics_cgate
python paper/make_figs.py
python paper/make_supp.py
python paper/scaffold_overlap.py
```

`-kl-coef 1.0` is mandatory for BRICS (the default 0.3 was validated for SELFIES only); runs are comparable only at a fixed thread count.

4 Corrections made to earlier claims

The repository records the following hypotheses that measurement overturned; the first group is listed in the repository's `limitations.md`, the last item concerns this manuscript.

1. "195/199 distinct samples, so no mode collapse": wrong metric; string uniqueness is blind for about 100 steps.
2. "The collapse is property-space, with only mild scaffold narrowing": true at step 40, over-generalized; at 200 steps scaffold collapse is larger.
3. "The reward uses the wrong classifier": overclaimed; no measurable out-of-distribution difference.
4. "Off-distribution the ensemble will disagree": backwards; it agrees more.
5. "Architecture differences are hopeless at 3 seeds": used marginal sds; the design is paired.
6. "The reward has gone blind": overstated; the information is there on a scale that could not show it.
7. Descriptor distance, applicability domain and Tox21 were each expected to catch the solvents; none does.
8. "The alert penalty makes the generator cleaner than approved drugs": true only on the instrument it was trained against (BRENK: 0.406 against drugs' 0.350).
9. BRICS "loses diversity because of its compact action space": retracted; the KL coefficient did not transfer (10.7$\times$ the KL).
10. The three-seed GAT-over-GIN effect did not replicate at ten seeds ($+0.0109$, CI $[-0.0057, +0.0275]$); the three-seed scaffold mean (0.864) was one unlucky draw (0.854 at ten).
11. A trioxide said to pass the screen did not (arm A's pool only); a ring-strain pattern that flagged 138 survivors was a labelling error and was removed.
12. *This manuscript:* Table 2 standard deviations for hybrid-BatchNorm (0.032, 0.047) and the 8-descriptor LightGBM value on B3DB (0.852 ± 0.040) were corrected against the result CSVs (the repository README table carries the earlier values); the claim that no calibration analysis exists was replaced by the measurement above; "36.8% share a training scaffold" referred to an older two-run survivor set and was recomputed (34.8% arm A, 23.5% gate arm, `scaffold_overlap.py`).

5 Files in the repository's `paper/` directory

`main.tex` and `supp.tex` (sources of the paper and this supplement), `figs/` (all figures), `make_figs.py` (all figures), `make_supp.py` (this document's data tables), `scaffold_overlap.py` (scaffold-overlap statistics), and `data/seed0_history.csv`, `data/seed1_history.csv`, `data/seed2_history.csv`: the original unconstrained-baseline histories from commit `e0e5da8b`, used for the collapse figure because `results/gan/seed*/history.csv` on the branch were later overwritten by a regenerated run.